

4 (a) The equation of state for an ideal gas may be written as

$$pVA = NBT$$

where  $p$  is the pressure of the gas,  $V$  is the volume of the gas,  $A$  is the Avogadro constant,  $B$  is another constant and  $N$  is the number of molecules of the gas.

(i) State the meaning, in the equation, of the symbol  $T$ .

..... [1]

(ii) Identify the constant  $B$ .

..... [1]

(b) The product  $pV$  for an ideal gas is also given by

$$pV = \frac{1}{3}Nm\langle c^2\rangle.$$

(i) State the meanings, in this equation, of the symbols  $m$  and  $\langle c^2\rangle$ .

$m$ : .....

$\langle c^2\rangle$ :.....

[2]

(ii) Use the equations in (a) and (b) to derive an expression, in terms of  $A$ ,  $B$  and  $T$ , for the mean kinetic energy  $E_K$  of a molecule of the gas.

$$E_K = ..... [2]$$

(c) On Fig. 4.1, sketch the variation with  $T$  of the root-mean-square (r.m.s.) speed of the molecules of an ideal gas.

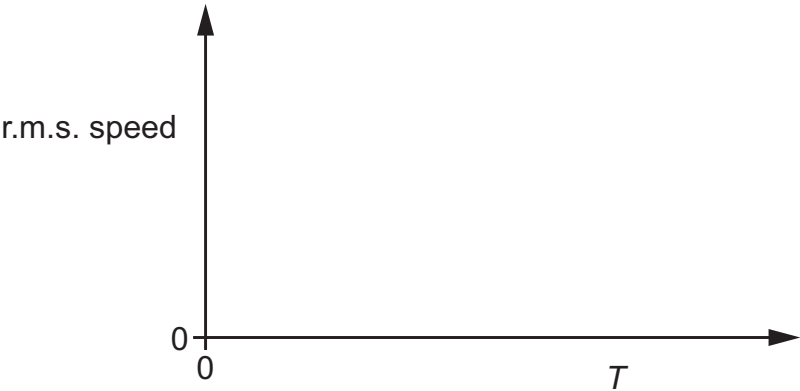


Fig. 4.1

[2]